

Human-in-the-Loop AI

Using Posit Assistant in Positron to Accelerate Analysis
Without Sacrificing Control

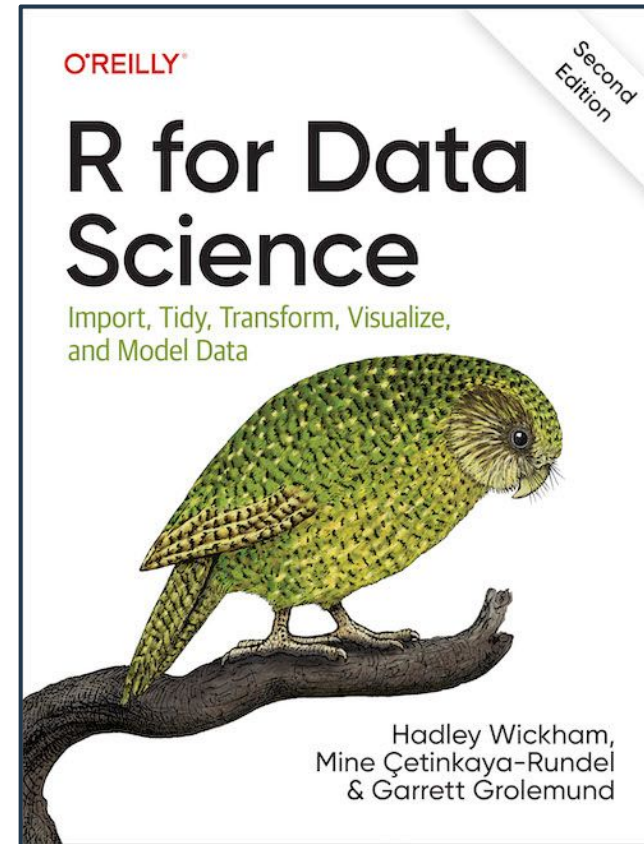
R/Pharma 2026



Your Instructor



Garrett Grolemond
Director of Learning
Posit PBC



Posit Academy

FREE

Self-paced, on-demand learning for every Posit user. No cost. Live now at academy.posit.co



Solutions

Products

Open Source

Customers

Resources

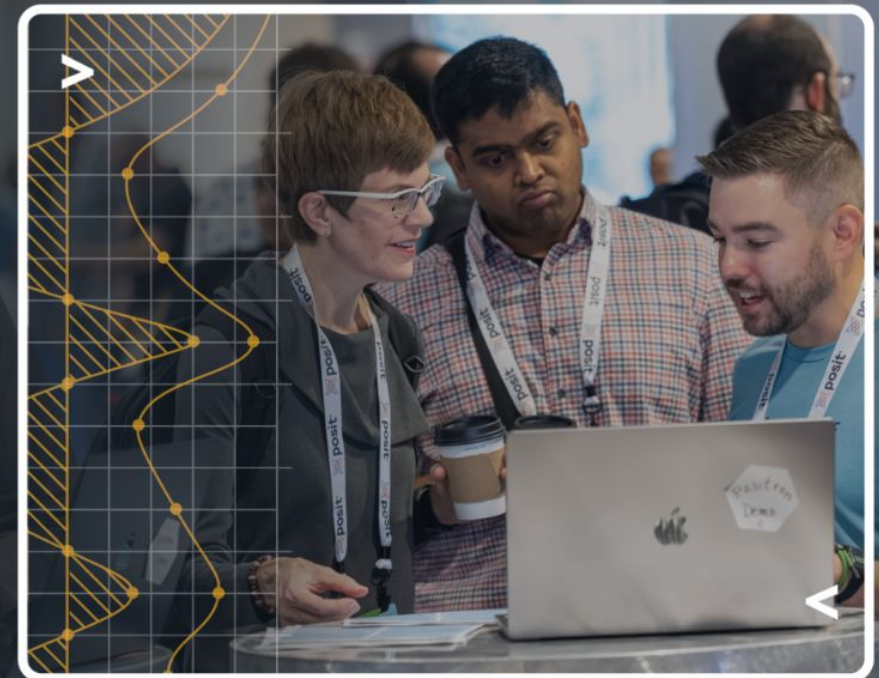
Search



Welcome to Posit Academy

Your learning hub for data science with R and Python — courses, live trainings, and a thriving community.

FULL COURSE CATALOG



Get Help

posit::conf(2026) is coming to Houston, TX! Join us Sept 14–16.

RSVP NOW

AI accelerates results, but what about?

- Security
- Validity
- Cost
- Transparency
- Reproducibility
- **Control**

Security

Make the model secure:

- HIPAA-compliant BAA with the model provider
- A local model in a compliant environment

Or make the data secure:

- Remove Personal Identifier Information
- No AI in environments with sensitive data

This is a bad idea.

An unsafe
LLM

Your IT
safeguards



AI accelerates results, but what about?

- Security
- Validity —
- Cost ?
- Transparency
- Reproducibility
- **Control**



Learning Objective

Enable every participant to begin using AI in Positron for pharma workflows.

Let's boot up our environment now!

workshop.posit.team

Do this with me.



Let's do this together

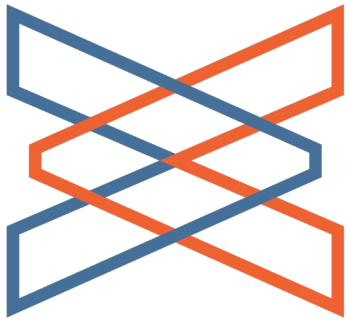
<https://workshop.posit.team>

1. Click on **Workbench**
2. Register for an account
3. Click **New Session**
4. Under **Image**, select **Workshop (2026-09-25)**



Quick Orientation to Positron





posit[®]
formerly **R** Studio[®]



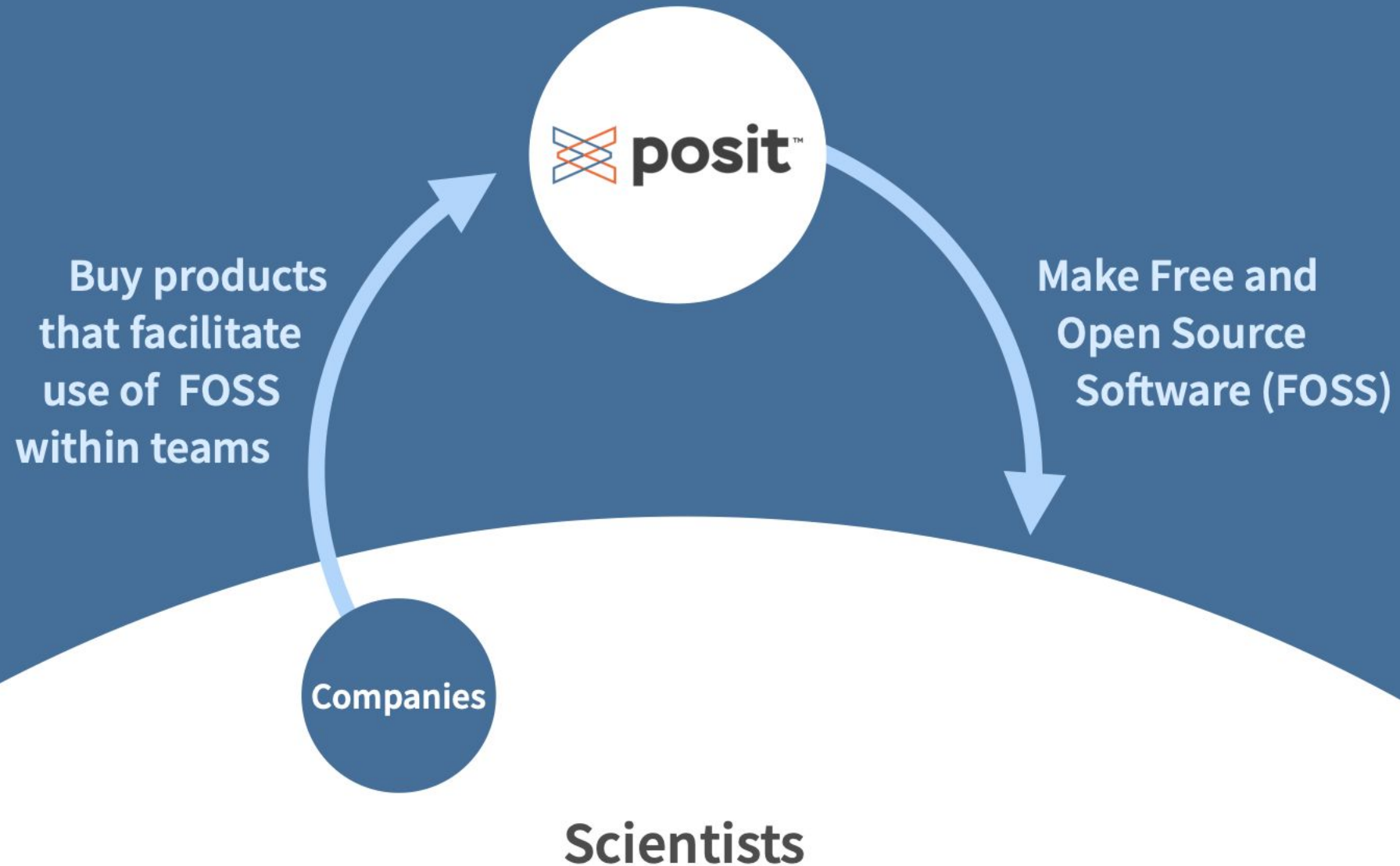
**OPEN
SOURCE
PLEDGE**
MEMBER



Posit's mission is to create free and open-source software for data science, scientific research, and technical communication, with the goal of enhancing the production and consumption of knowledge for everyone, regardless of economic means.

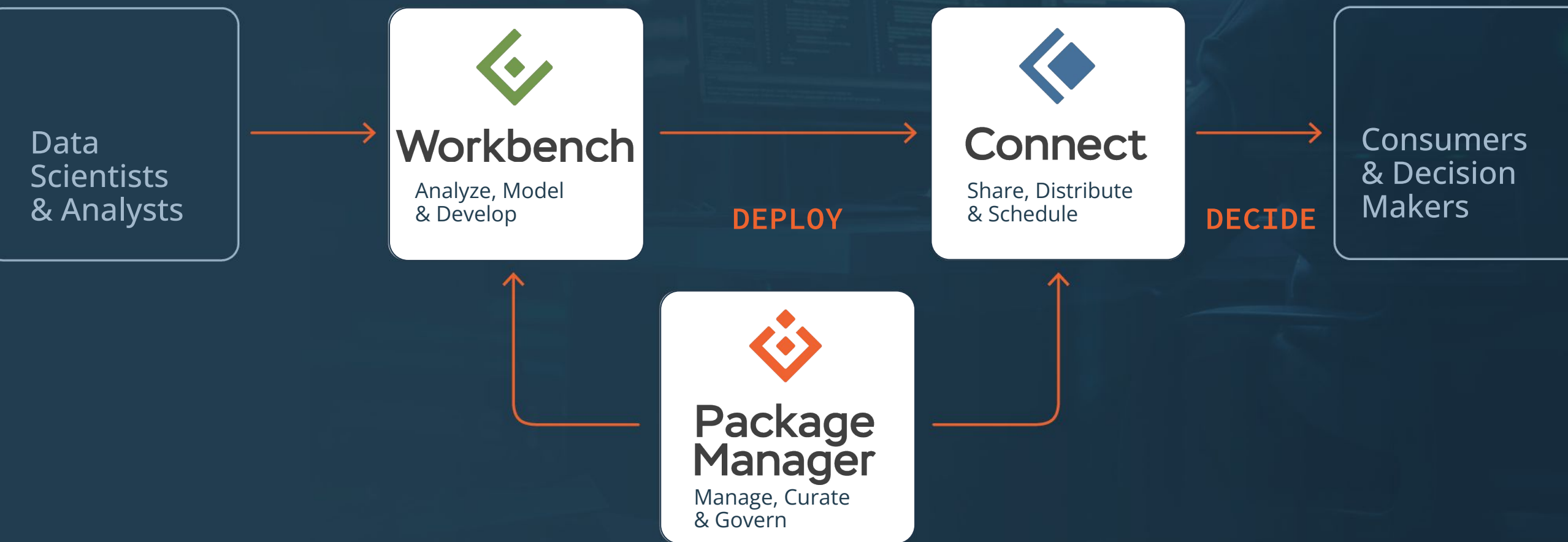


The Virtuous Cycle

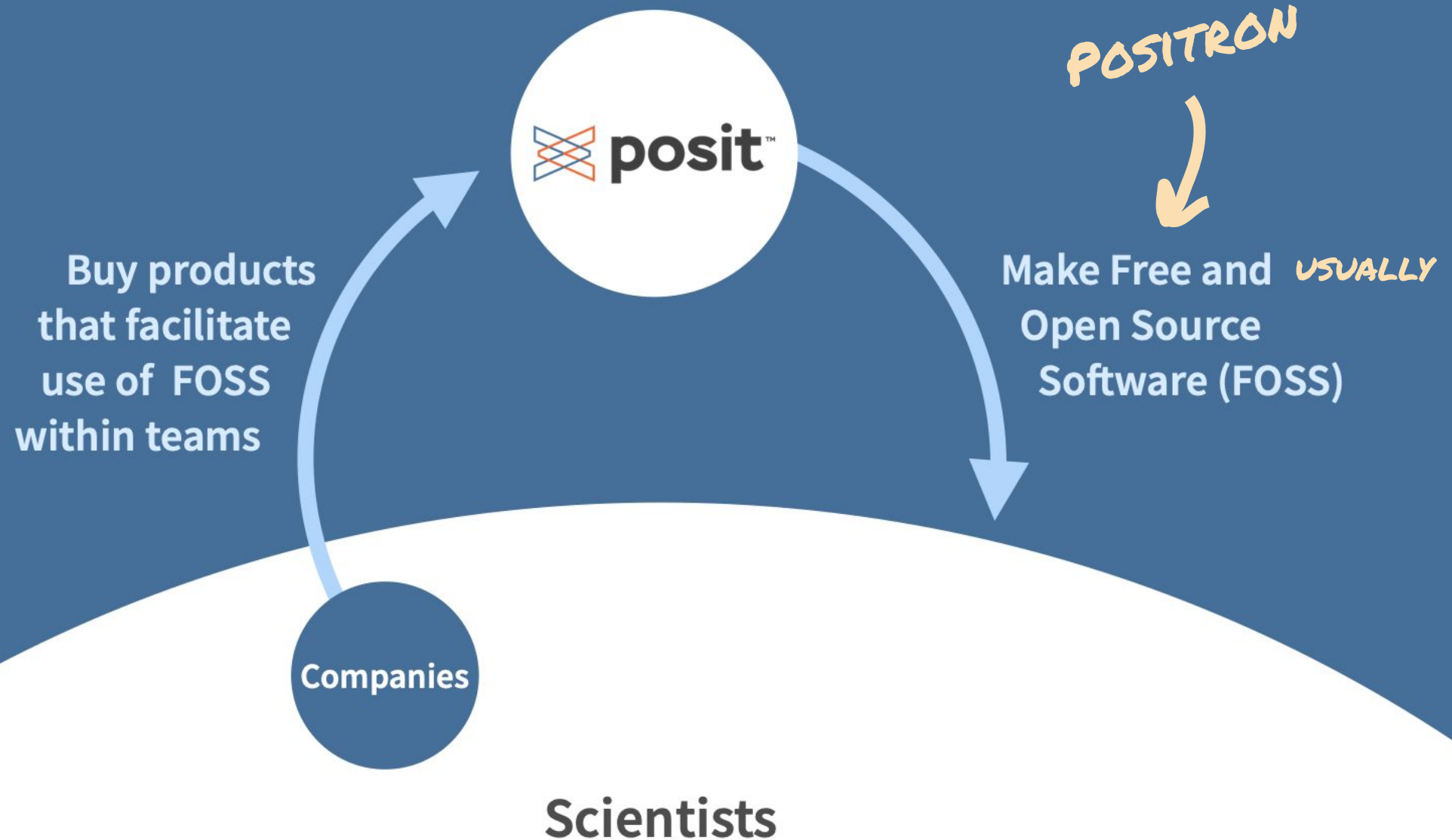


Posit Team - A Professional Insight Suite

A complete intelligence stack for code-first data science and research. Data scientists can work safely, independently, and quickly, without IT handoffs or hand-holding.



The Virtuous Cycle



Your Turn

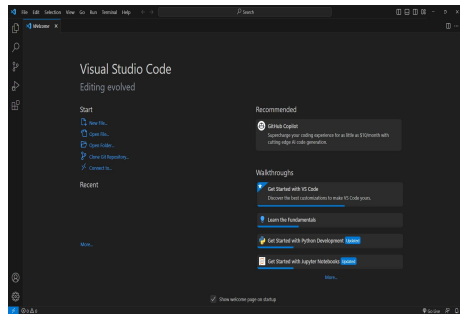
Poll - Which IDEs (Code Editors) have you used?

What is Positron ?

A free fork of Visual Studio Code purpose-built for **data science**

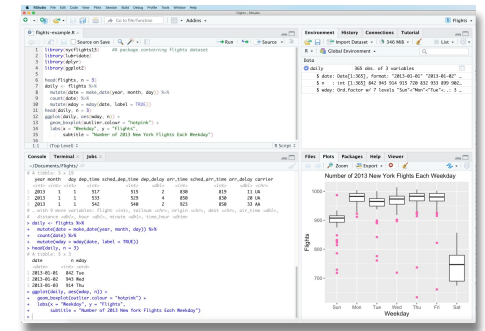
POSITRON

VS CODE

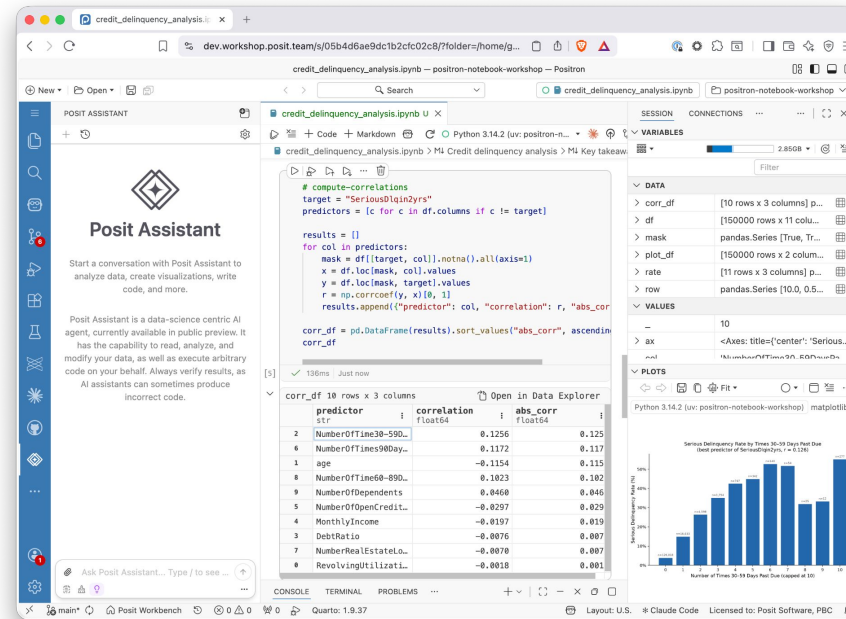


Software Developers

RSTUDIO

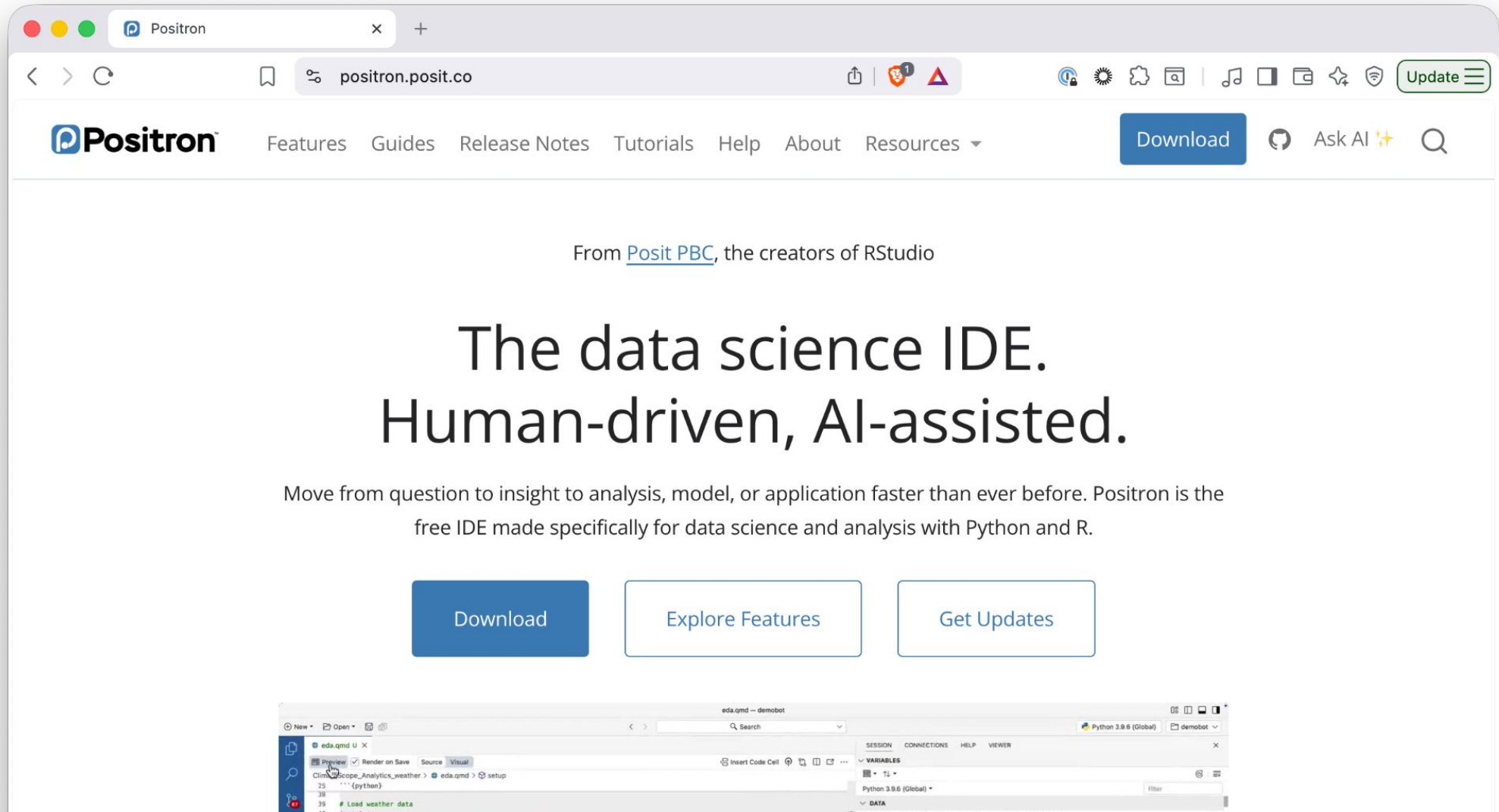


Data Scientists



Data Scientists

positron.posit.co



Your Turn

Poll - Which language(s) do you use?

Today's materials

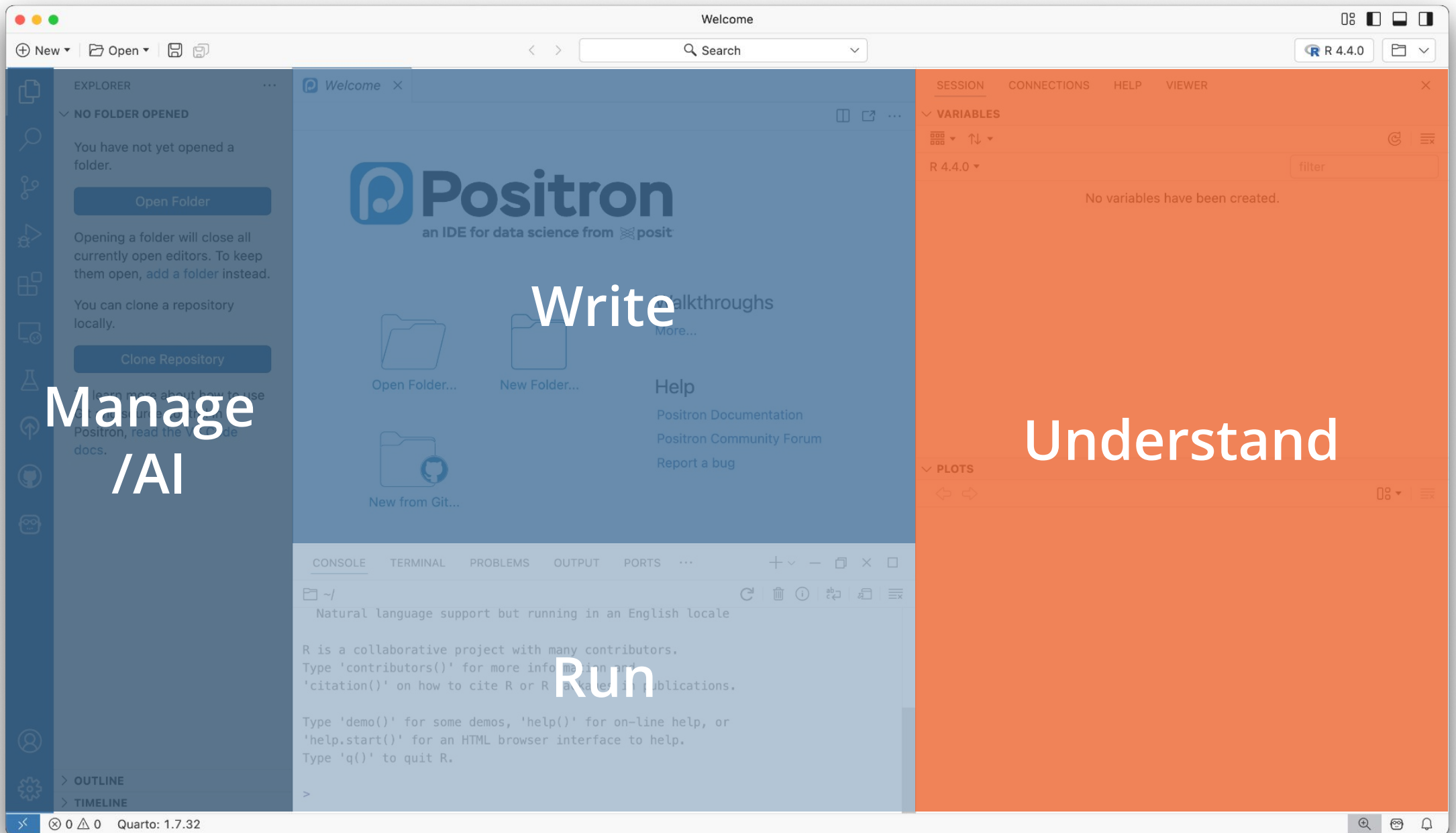


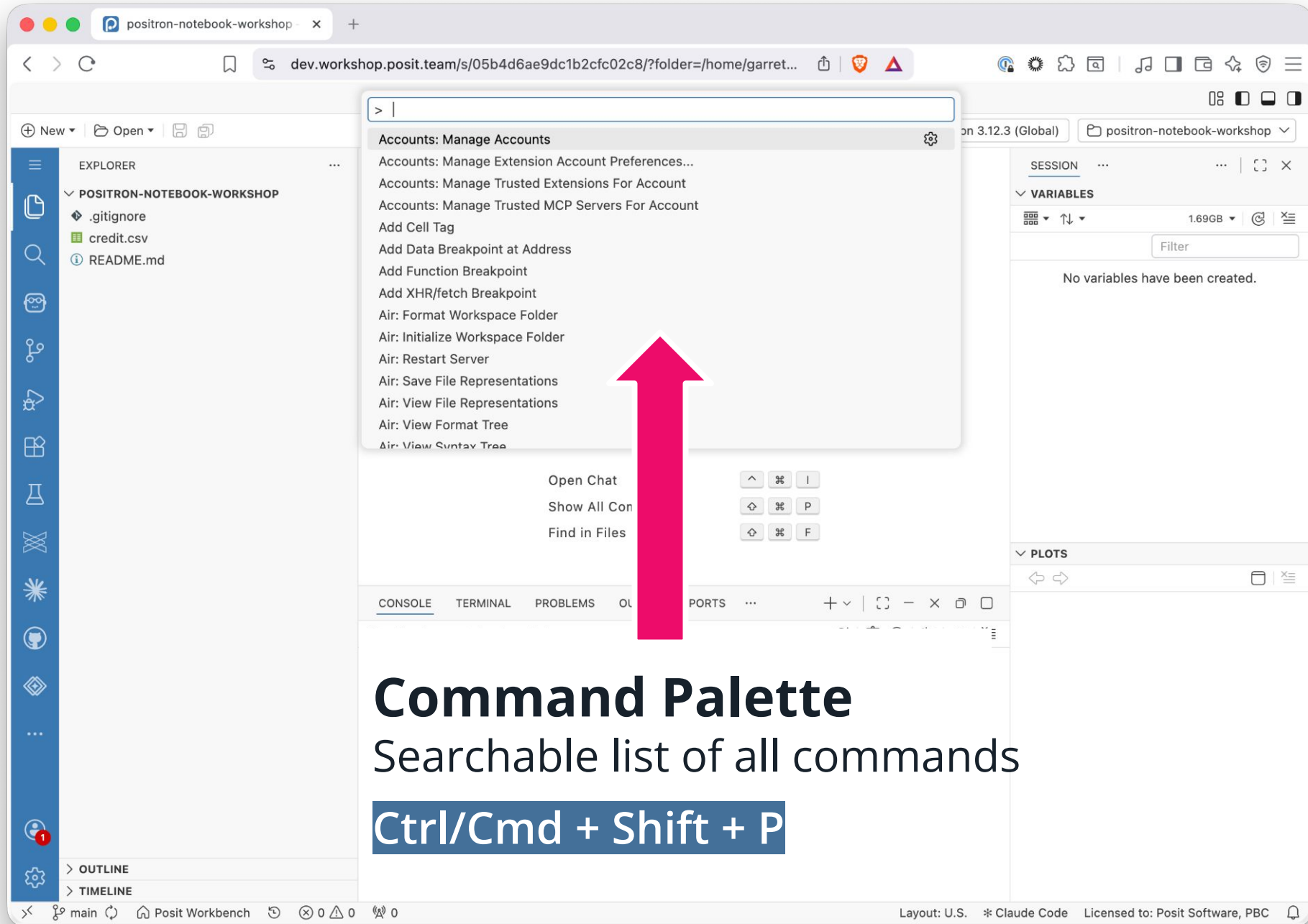
Will remain available after the workshop at
<https://github.com/posit-dev/human-in-the-loop-ai-r-pharma-2026>



Demo - Set up and tour of Positron







positron-notebook-workshop — Positron

Python 3.12.3 (Global)

Search

EXTENSIONS

Search Extensions in Marketplace

DEV.WORKSHOP.POSIT.TEAM - INSTALL... 17

- Air - R Language Support**
R formatter and language server
Posit Software, PBC
- Claude Code for VS C...** 802ms
Claude Code for VS Code: Harnes...
Anthropic
- Even Better TOML**
Fully-featured TOML support
tamasfe
- GitHub Pull Requests** 893ms
Pull Request and Issue Provider f...
GitHub
- Jupyter** 193ms
Jupyter notebook support, intera...
ms-toolsai
- Jupyter Cell Tags**
Jupyter Cell Tags support for VS ...
ms-toolsai
- Jupyter Keymap**
Jupyter keymaps for notebooks
ms-toolsai
- Jupyter Notebook Rend...** 2ms
Renderers for Jupyter Notebooks ...
ms-toolsai
- Jupyter Slide Show**
Jupyter Slide Show support for V...

BROWSER - INSTALLED 0

RECOMMENDED 0

Layouts

Hide everything but the source pane

SESSION ...

VARIABLES

No v... e been created.

PLOTS

CONSOLE

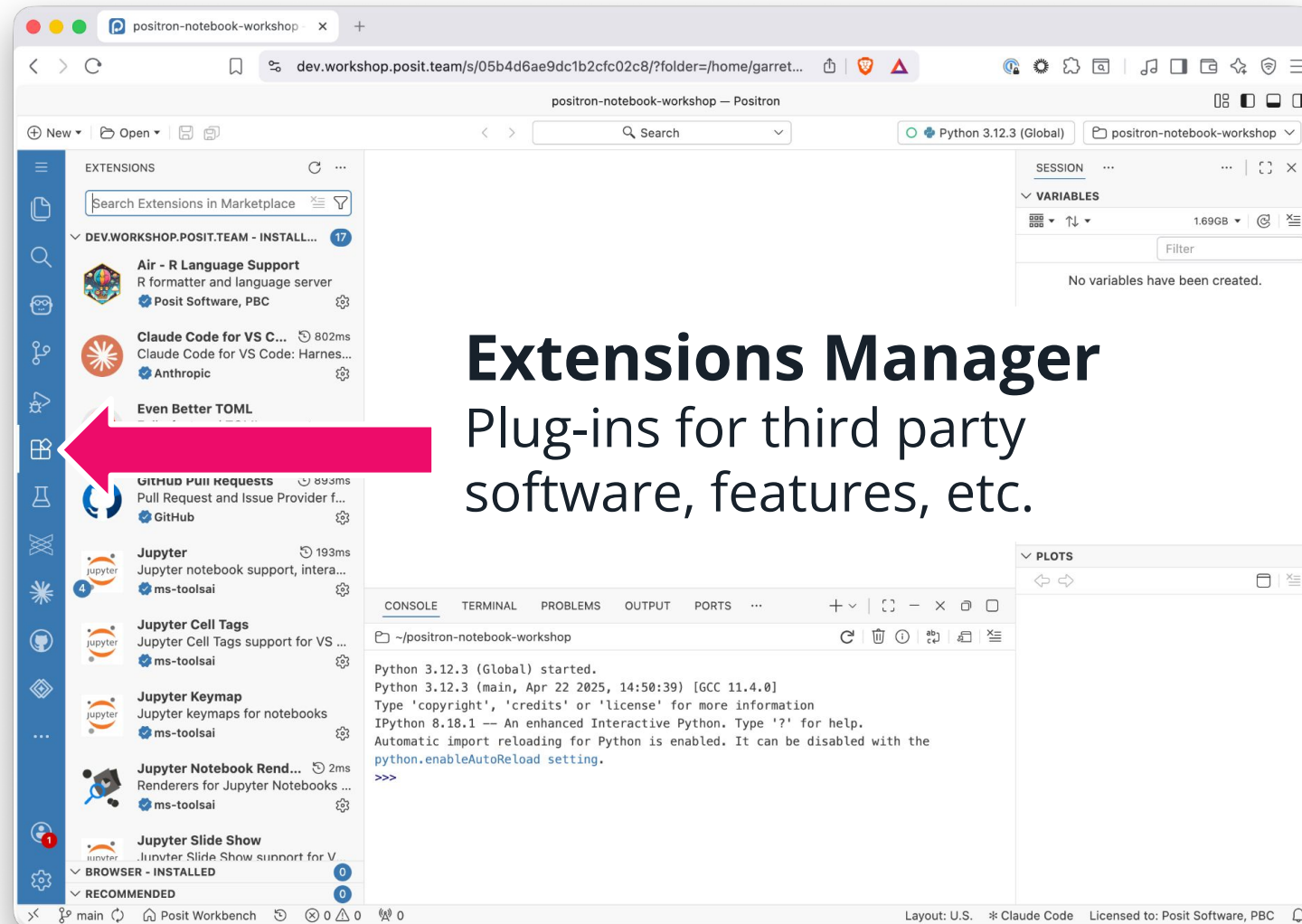
~/positron-notebook-workshop

Python 3.12.3 (Global) started.
Python 3.12.3 (main, Apr 22 2025, 14:50:39) [GCC 11.4.0]
Type 'copyright', 'credits' or 'license' for more information
IPython 8.18.1 -- An enhanced Interactive Python. Type '?' for help.
Automatic import reloading for Python is enabled. It can be disabled with the [python.enableAutoReload](#) setting.
>>>

main Posit Workbench 0 0 0 0

Layout: U.S. * Claude Code Licensed to: Posit Software, PBC

Extensions for SAS users



Your Turn

Open the Command Palette with
Cmd/Ctrl + Shift + P.

Find and run the command
Developer: Reload Window.

Your Turn

Open the Command Palette with
Cmd/Ctrl + Shift + P.

Find and run the command
Developer: Reload Window.

AI with Positron





Posit Assistant

An
AI client
within Positron.



Adapts *your LLMs* to data science. It:



1. Provides data science skills



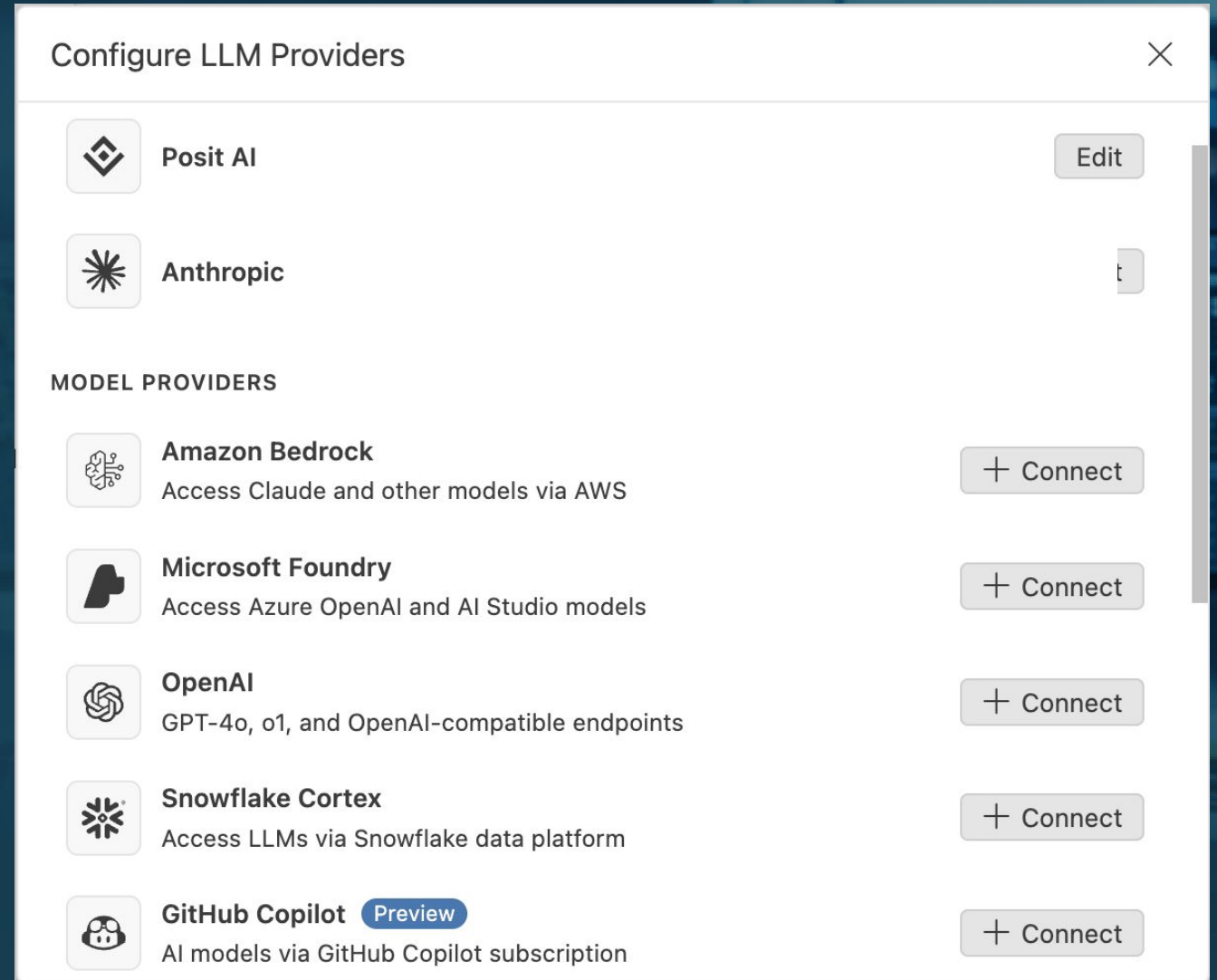
2. Encourages a code-first, token saving method



3. Organizes access to Positron, R, and Python sessions



Power Positron with your preferred LLM



Power Positron with your preferred LLM

Configure Language Model Providers

Provider


Posit AI


Amazon Bedrock


Anthropic


Microsoft Foundry


OpenAI


Snowflake Cortex


GitLab Copilot
Pre

Authentication

☒ OAuth ☐ API Key

[Sign in](#)

Get started with Posit Assistant instantly via a free trial of [Posit AI](#), a managed service that provides access to frontier LLMs through a single account. Posit AI provides access to both Posit Assistant and Next Edit Suggestions.

By using Posit AI, you agree to the [Posit EULA](#), [Posit AI Terms of Service](#), and [Privacy Policy](#).

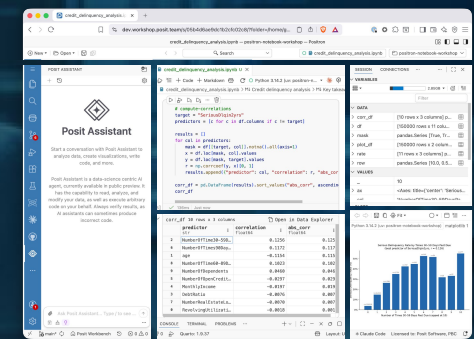
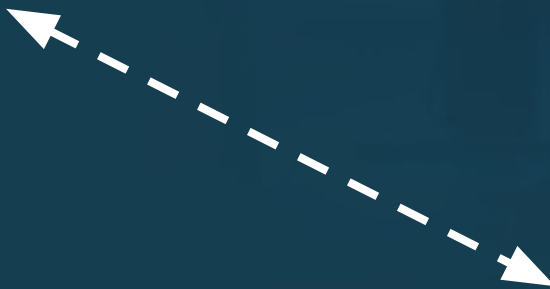
Your use of Posit AI is optional and at your sole risk.

[Close](#)

AI

You

Files



Quiz - Does this code return an error?

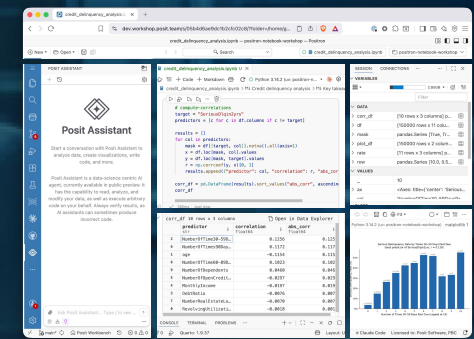
```
hist(cases$amount)
```





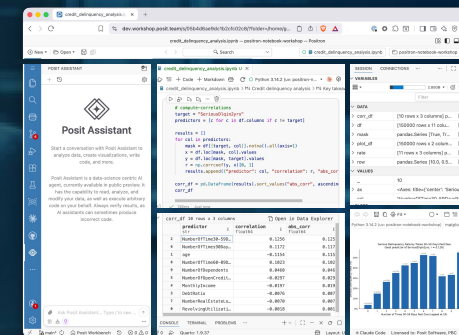
Posit Assistant

AI





?



You



AI





Demo - Open and configure Posit Assistant



Upskill your models **and** bring them into your session

Posit Assistant uses your environment as context, so its suggestions **work immediately** with your data, plots, tables, and code.

The screenshot displays the Posit Assistant interface within a Quarto document. The interface is divided into several panels:

- Left Panel:** Contains the Posit Assistant logo and a vertical toolbar with icons for file operations, search, and other tools.
- Central Panel:** Shows the code editor with R code for generating a plot. The code includes comments and function calls like `geom_map` and `plot_per_capita_spend`. Orange arrows point from the Posit Assistant logo to specific lines of code, indicating suggestions or context-aware assistance.
- Right Panel:** Displays the environment and data context, including a table of variables and a plot titled "Income". The plot shows a bar chart of per-capita spending by income group.
- Bottom Panel:** Shows the terminal and status bar, indicating the current session is using R 4.6.0 and Quarto 1.9.36.

The Posit Assistant logo, a stylized diamond shape, is prominently displayed in the center-left of the interface, with orange arrows pointing to the code editor and the plot, illustrating its role in providing context-aware suggestions and assistance.

Follow along with me

Open Posit Assistant ( in sidebar). Ask it to diagnose a double programming discrepancy:

ADAE/ADAE_prod.csv and ADAE/ADAE_qc.csv are two versions of an ADAE extract. They should be identical but are not. I'd like to figure out the cause. Find every discrepancy, and tell me if there's a pattern to which records disagree.

Follow along with me

Open Posit Assistant ( in sidebar). Ask it to diagnose a double programming discrepancy:

ADAE/ADAE_prod.csv and ADAE/ADAE_qc.csv are two versions of an ADAE extract. They should be identical but are not. I'd like to figure out the cause. Find every discrepancy, and tell me if there's a pattern to which records disagree.

You



AI



Posit Assistant

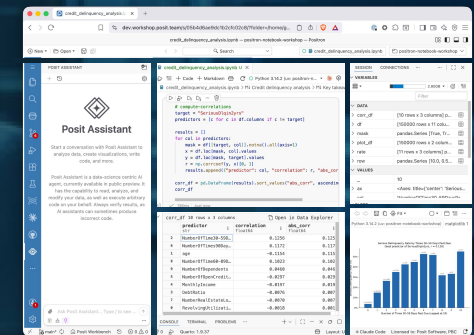
Skills, context



results

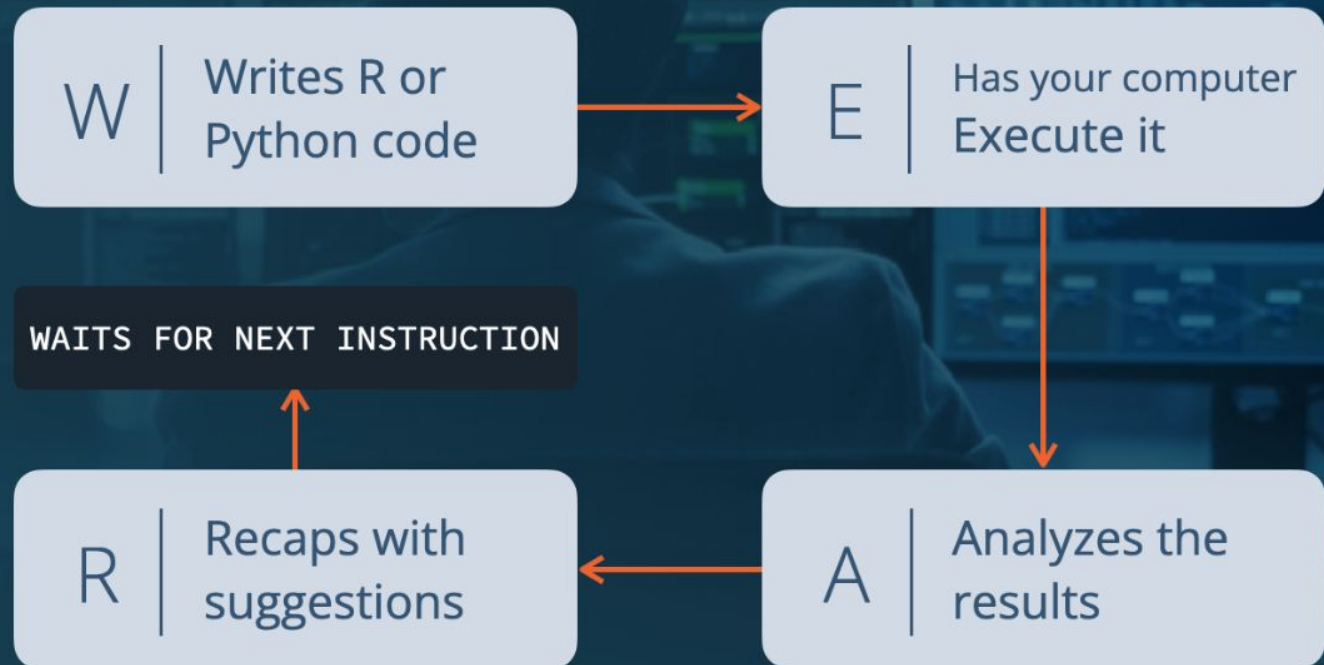


code



The
WEAR
philosophy

Code is the **most auditable**
way to explore data with AI



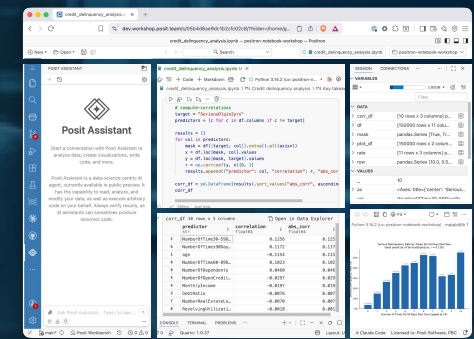
You



AI



Posit Assistant



 **executeCode:** load both ADAE extracts and compare structure

Can Posit Assistant use executeCode?

Posit Assistant would like to run this R code to load both ADAE extracts and compare structure.

```
library(tidyverse)

prod <- read_csv("ADAE/ADAE_prod.csv", show_col_types = FALSE)
qc <- read_csv("ADAE/ADAE_qc.csv", show_col_types = FALSE)

list(
  dim_prod = dim(prod),
  dim_qc = dim(qc)
)
```

✓ Allow

✗ Decline



Allow once

✓ Allow executeCode tools for this session
for all inputs

📁 Always allow executeCode tools for this
project
for inputs matching *

What should Posit Assistant do instead?

Ex: Please use a different plotting library.

Cancel

Decline

Allow, decline,
and edit your
LLM code requests

Working with AI



Test Drive

Who wants to take the wheel?

Your Turn

Click + (top left) to start a new conversation.

Then ask Posit Assistant to build an ADSL table:

Build an ADSL table from the SDTM files in SDTM/ using the specification in specs/ADSL_spec.xlsx. Use {metacore} to read the spec, {admiral} to derive the variables it describes, {metatools} to check the result against spec, and {xportr} to export a submission-ready .xpt.

Follow up as needed.

Your Turn

Click + (top left) to start a new conversation.

Then ask Posit Assistant to build an ADSL table:

Build an ADSL table from the SDTM files in SDTM/ using the specification in specs/ADSL_spec.xlsx. Use {metacore} to read the spec, {admiral} to derive the variables it describes, {metatools} to check the result against spec, and {xportr} to export a submission-ready .xpt.

Follow up as needed.

Collecting results





Summarize your conversation
with the **/notebook** and
/report commands

Commands

/report Create a report as a Quarto document

/notebook Create a report as a Jupyter notebook

/



Claude 4 Sonnet ▾



Your Turn

Run `/report` in Positron Assistant to save your code in an editable format.

Then click **Preview** at the top of the .qmd file

Your Turn

Run `/report` in Positron Assistant to save your code in an editable format.

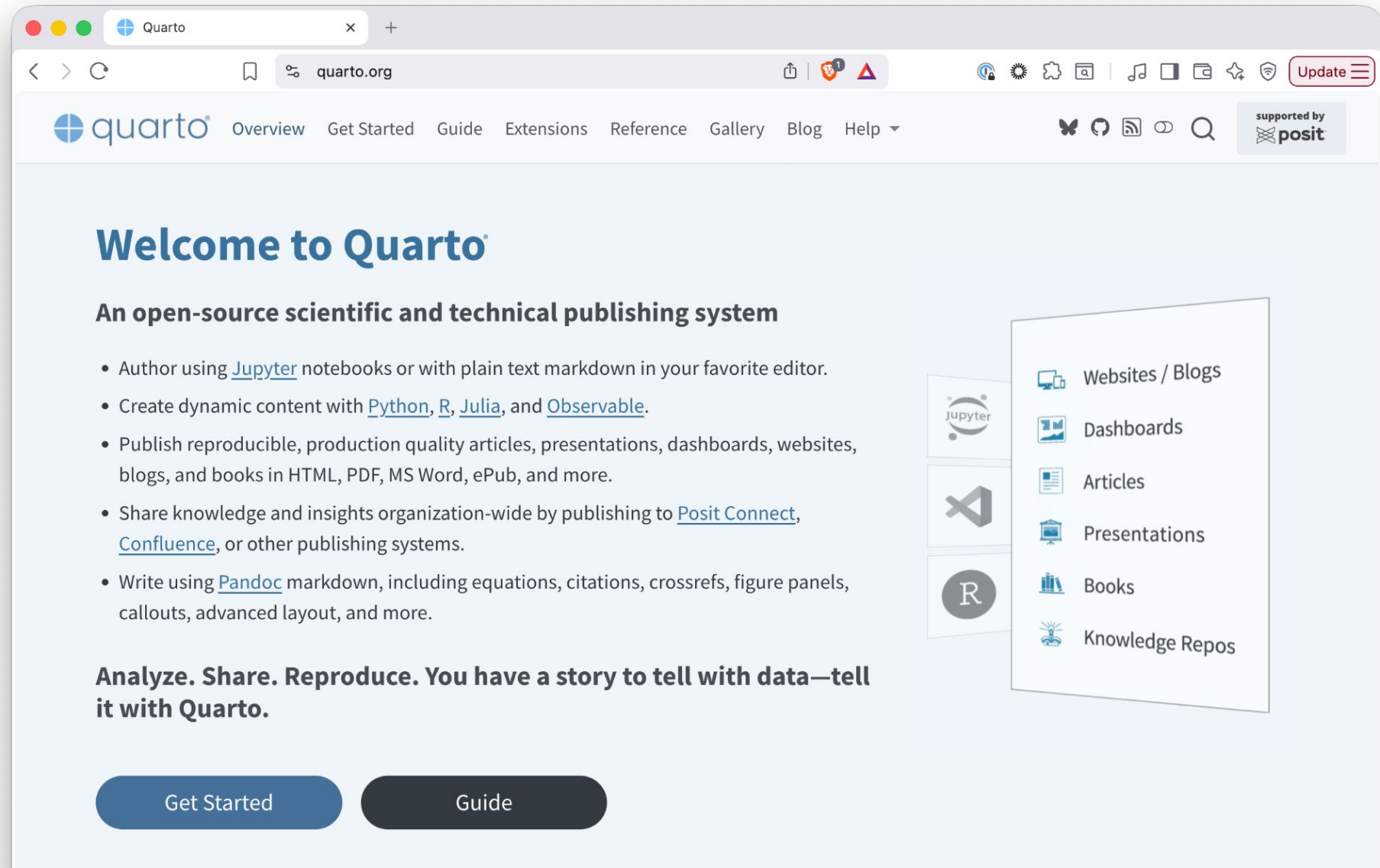
Then click **Preview** at the top of the .qmd file

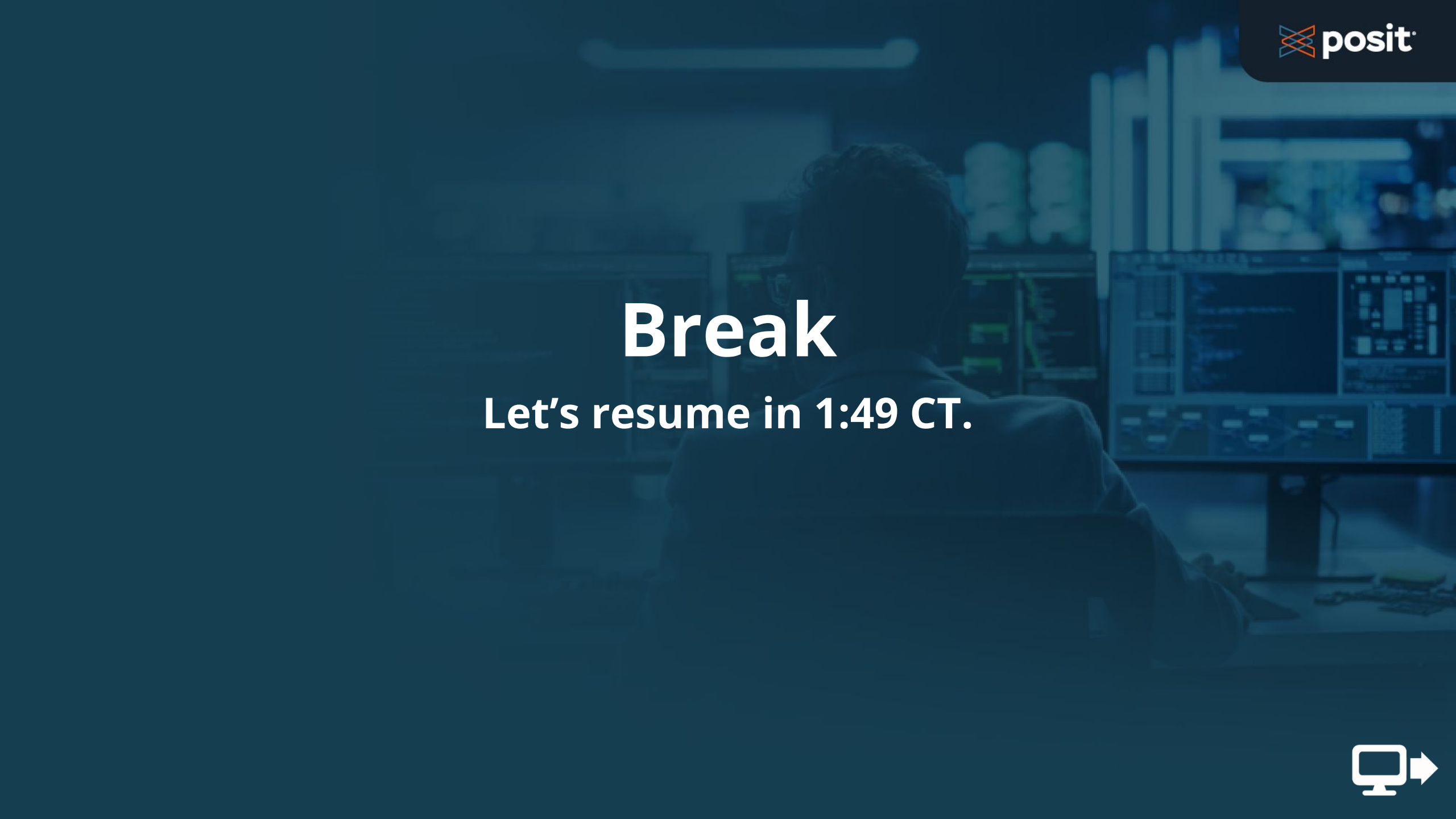


Demo - Quarto reports for R and Python.



quarto.org



The background is a dark, blue-tinted image of a person sitting at a desk with multiple computer monitors. The person is seen from the back, looking at the screens. The monitors display various data visualizations, including charts and graphs. The overall atmosphere is professional and tech-oriented.

Break

Let's resume in 1:49 CT.



Monitoring tokens

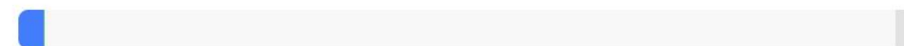


Posit Assistant helps you **manage your token usage** and keep it **transparent**.

Know **how many tokens** you are using and what you are using them for. Estimate the cost, and keep your cache alive between queries.

Session information

Context



● System ● Tools ● Chat

27,793 / 1,000,000 tokens (2.8%) · Compacts at 970,000

Token Usage

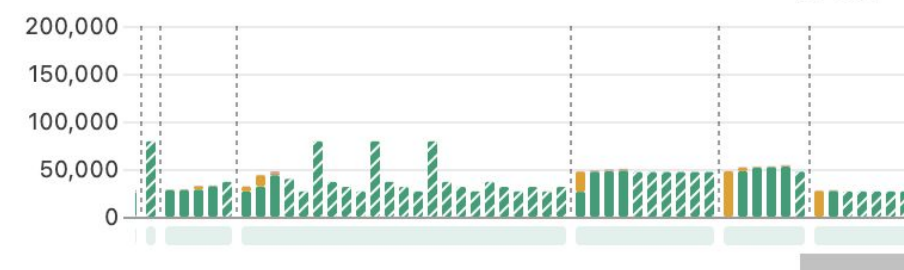
Conversation

Session

Cost

Tokens

CSV



Keepalive

	Tokens	Cost
● Output	158,057	\$1.739
● Input (uncached)	540	\$0.001
● Input (cache write 5m)	707,939	\$1.947
● Input (cache read)	11,100,931	\$2.442
Keepalive rings ~ 68	2,012,388	\$0.670

799



Ask Posit Assistant...



1:32



27K · \$0.12



Sonnet 5 (Medium)



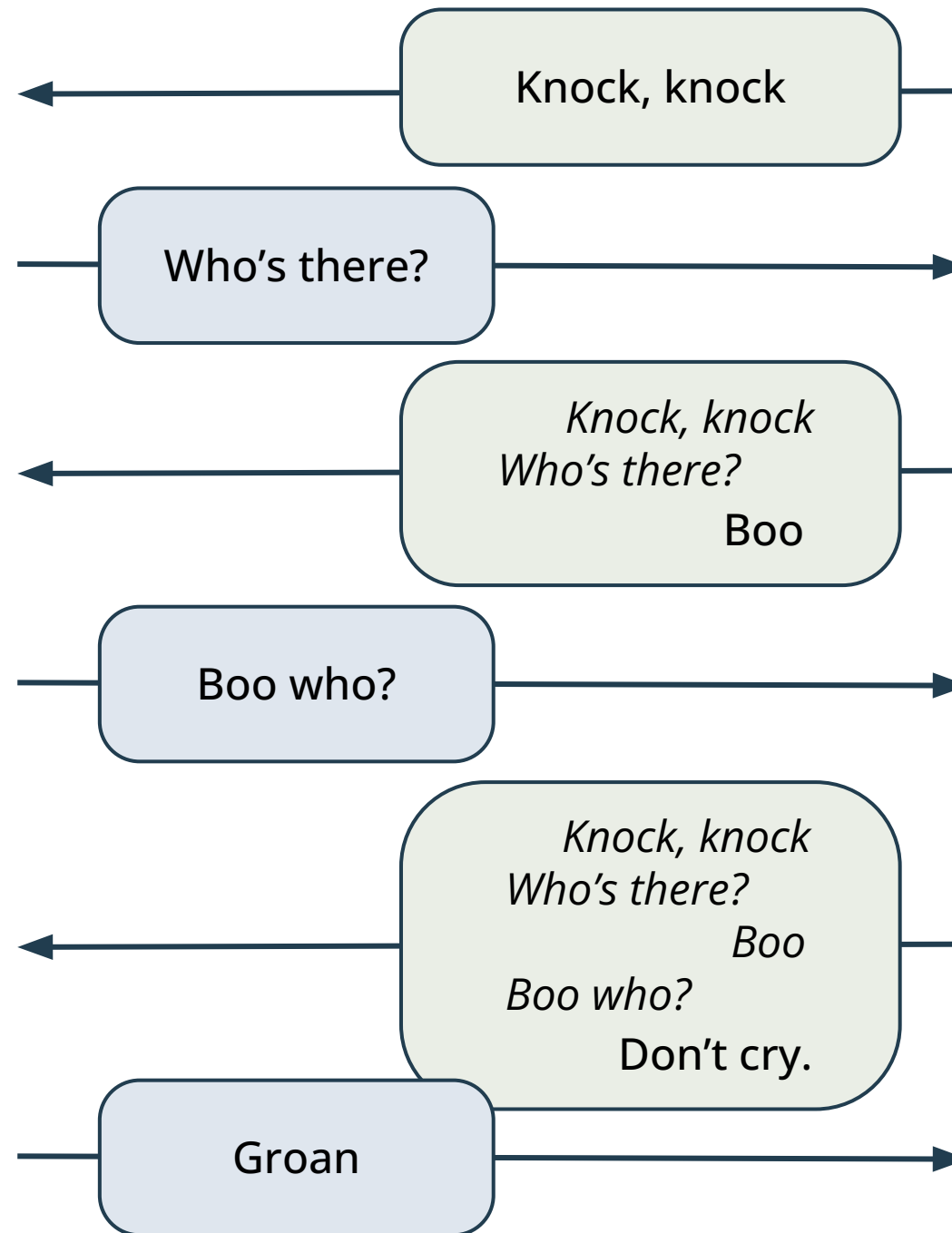
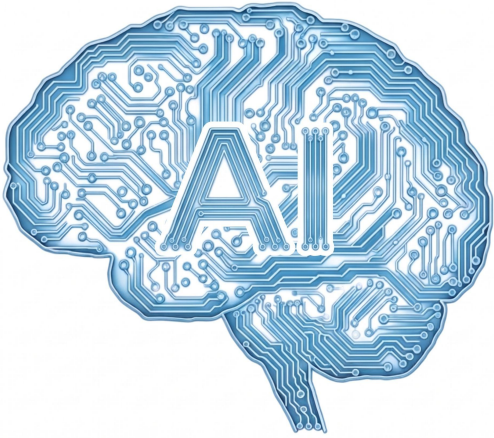
☒ Show the cost of each turn in its footer ⓘ 🗑️

Prompt cache



Demo - Monitoring token usage in Posit Assistant, and working with AGENTS.md and skills.





Tips

For minimizing token usage:

1. **Model** - Select the smallest one for the job
2. **Cache** - Keep the cache alive if you plan to do more soon
3. **Prompts** - Write clear prompts

Prompt Advice

For each prompt, explain:

1. **Why** - Explain your ultimate goal
2. **What** - Clear instructions for what you want it to do
3. **How** - Which functions, files, and *packages* you want it to use

I'd like to use the combined happiness dataset to understand how the USA's happiness score and ranking changed from 2015 to 2019. **Build a plot and a table to illustrate the changes.** Use the ggplot2 and gt packages to do this.

Your Turn

Start a new conversation.

ADAE/AE_table_rtables.R creates a working AE frequency table built with {rtables}. Ask Posit Assistant to write an R script that rebuilds the exact same table, but instead uses {cards} and {tfrmt}.



Your Turn

Start a new conversation.

ADAE/AE_table_rtables.R creates a working AE frequency table built with {rtables}. Ask Posit Assistant to write an R script that rebuilds the exact same table, but instead uses {cards} and {tfrmt}.

Teaching the AI



Your Turn

Start a new conversation and ask Posit Assistant to make a KM plot.

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM. For this task, ignore the style guide in house_style.md.



Your Turn

Start a new conversation and ask Posit Assistant to make a KM plot.

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM. For this task, ignore the style guide in house_style.md.

How do you teach an LLM to do things a certain way?

Memory (AGENTS.md)

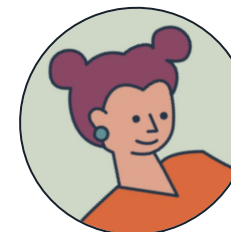
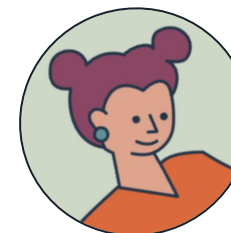
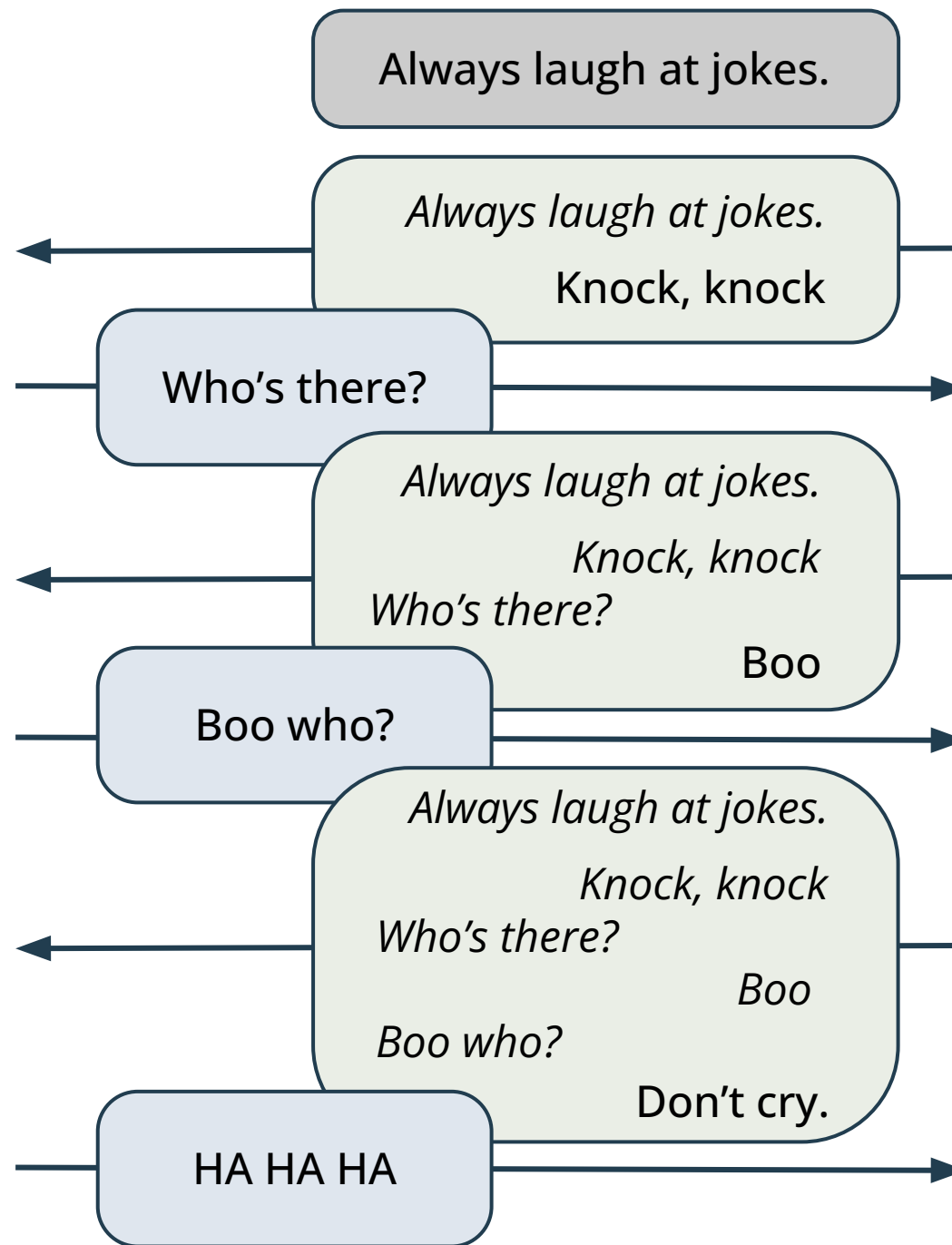
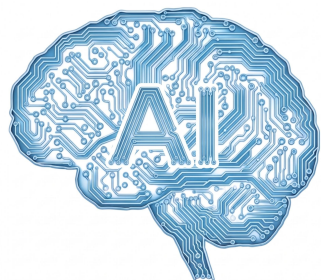
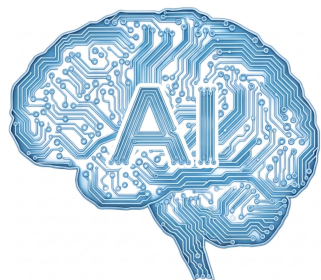
<https://assistant.posit.co/docs/features/memory/>

Skills

<https://assistant.posit.co/docs/features/skills/>

<https://github.com/posit-dev/skills>

AGENTS.md



Your Turn

Rename house_style.md to AGENTS.md.

Start a new conversation.

Run this prompt that does not mention style.

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM.

Your Turn

Rename house_style.md to AGENTS.md.

Start a new conversation.

Run this prompt that does not mention style.

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM.

AGENTS.md

When you make **plots**, I want you to follow these instructions.
(Several lines of instructions that get loaded into every conversation).

When you make **tables**, I want you to follow these instructions.
(Several lines of instructions that get loaded into every...

Hi

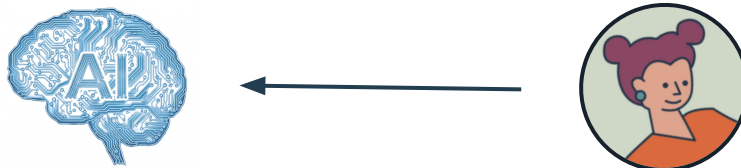


MCP

There are three tools you can use:

- **The Plot tool**
What the plot tool does. How to use the plot tool. What the plot tool returns. This takes many lines that get loaded into every conversation.
- **The Table tool**
What the tool...

Hi



Skills

These are available to read if you need them:

- How to make **plots**
- How to make **tables**

Hi





Create a skill with **/create-skill**

Slash Commands

- /compact** Compact conversation to reduce toke...
- /context** Show session context and token usage
- /clear** Start a new conversation, keeping the cur...
- /clean** Enter Cleaning mode
- /create-skill** Create new Agent Skills following t...
- /marketplace** Open the plugin & marketplace m...
- /mcp** Show MCP servers and sign in to them
- /microcompact** Micro-compact tool results in t...

/



Claude 4 Sonnet ▾



Your Turn

Run

/create-skill Use the information in AGENTS.md to build a skill that makes Kaplan Meier plots that match the described style. Then delete AGENTS.md.

Then, in a new conversation re-run

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM.

Your Turn

Run

/create-skill Use the information in AGENTS.md to build a skill that makes Kaplan Meier plots that match the described style

Then, in a new conversation re-run

Using ADAM/ADAE.csv and ADAM/ADSL.csv, derive the time to first treatment-emergent adverse event for the safety population and plot a Kaplan-Meier curve by ARM.

Posit makes and shares skills



A collection of skills from Posit

<https://github.com/posit-dev/skills>

What else is there?

MCPs, hooks, RAG,...

<https://assistant.posit.co/>

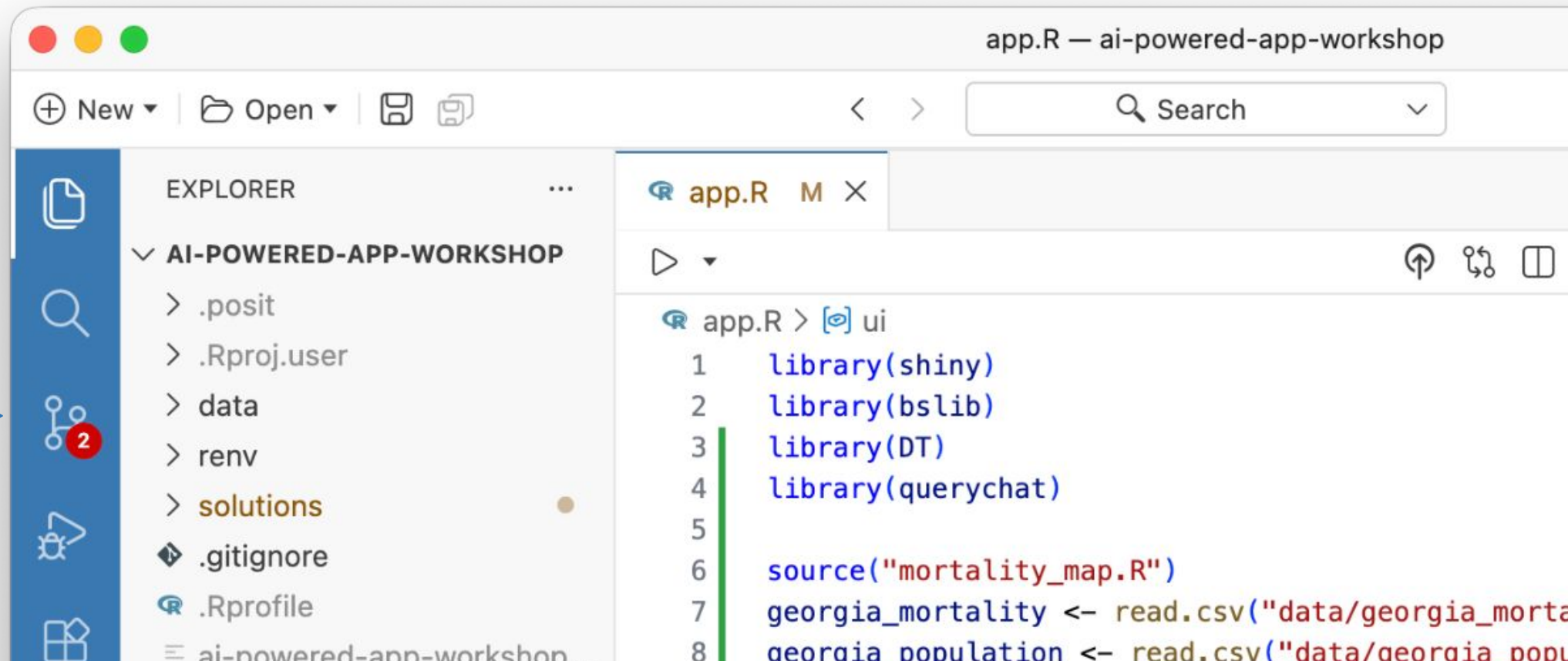
Publishing from Positron



Publish to Posit Connect



Push
to source
control for
Git-backed
publishing



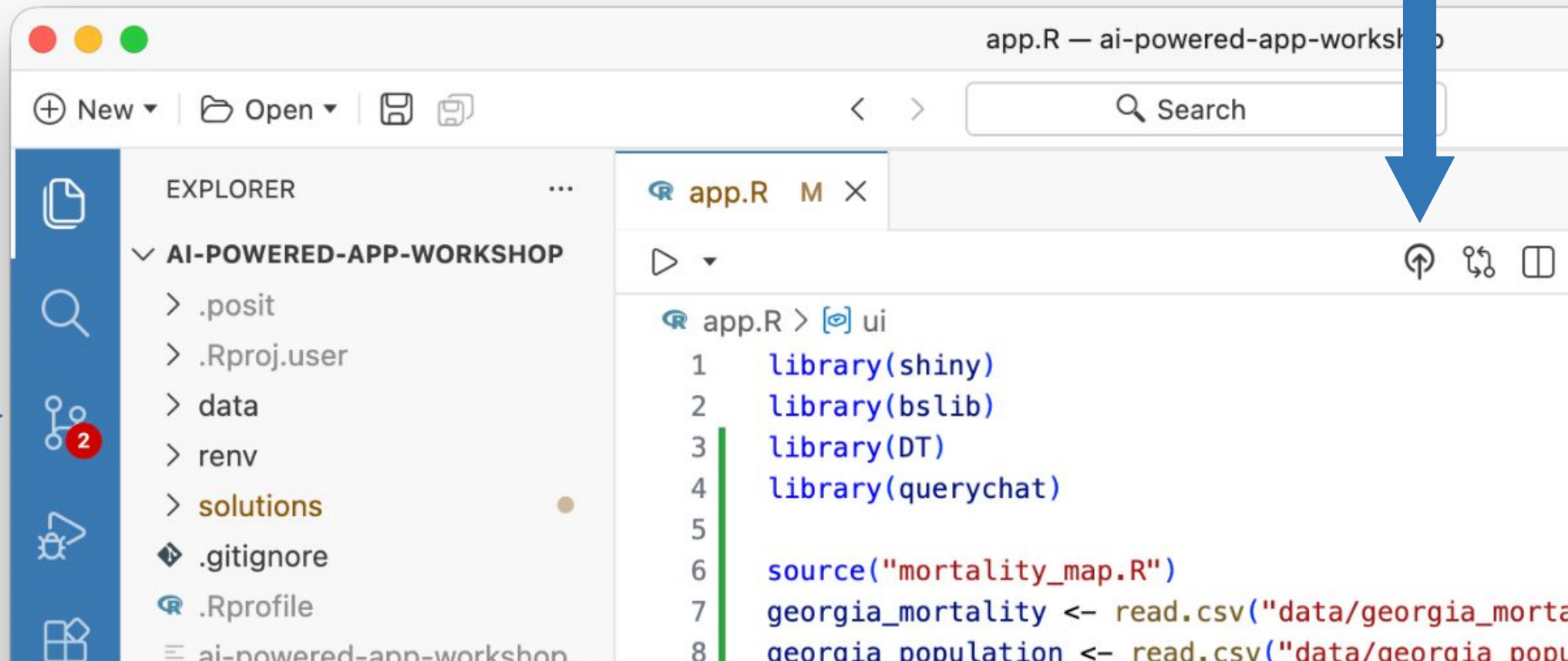
Publish to Posit Connect



Push button
publishing to
Posit Connect

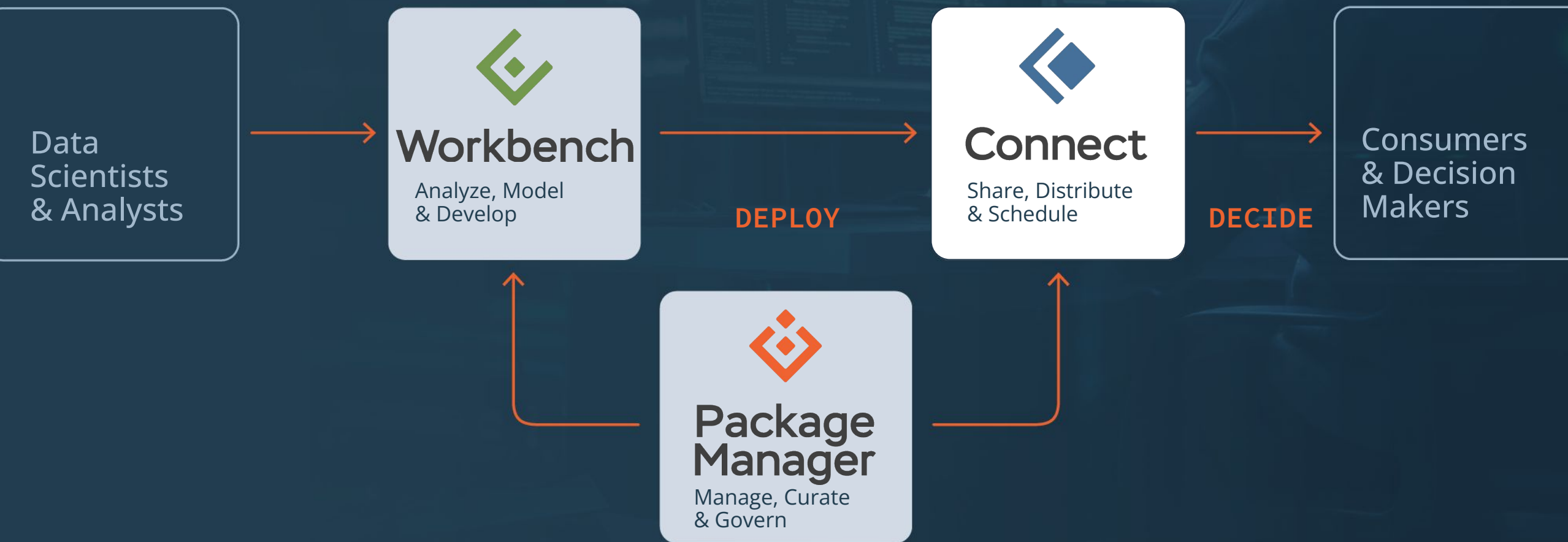


Push
to source
control for
Git-backed
publishing



Posit Connect is a part of Posit Team

Posit Connect makes it easy to deploy and share data science work across teams. Publish R and Python models, dashboards, APIs, and reports to the people who need insights most—securely, seamlessly, and at enterprise scale.





Demo - Using Git within Positron



Your Turn

Follow along with me to publish your document to Posit Connect.





Demo - Posit Connect features and Git-backed deployment



Recap



Human-in-the-loop AI with **Positron**

1. Decide which models you can use securely
2. Access them within your workflow through Posit Assistant
3. Use LLMs to generate code trails (WEAR loop + /report)
4. Keep track of token use
5. Make use of AGENTS.md, skills, etc.

Posit Academy

FREE

Self-paced, on-demand learning for every Posit user. No cost. Live now at academy.posit.co



ON-DEMAND

Product Courses

Self-paced courses covering Connect, Workbench, and Package Manager workflows. Available 24/7



HANDS-ON

Labs

Sandboxed Instruqt environments to practice real Posit workflows. No setup, no risk to production data: perfect for new users.



LIVE

Workshops

Live online sessions led by the CAPE team. Small-group, interactive format for customers who want guided practice.

FREE

Posit Academy

- On-demand courses, labs & live workshops
- Open to all customers, prospects & open source users
- Login and Learn

PAID

Academy Apprenticeships

- Cohort-based, mentor-led 6-8 week programs
- Premium upskilling in R & Python
- The same highly rated program with a new name

Please email me if you have an
example you would like to see
taught on academy.posit.co

garrett@posit.co

Thank you
What questions do you have?

Posit Academy

FREE

Self-paced, on-demand learning for every Posit user. No cost. Live now at academy.posit.co



Solutions

Products

Open Source

Customers

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